

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1959

Barling 138 kV
January 22, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1959. The Interconnection Customer proposes a 79.9 MW solar photovoltaic facility interconnecting at the Barling 138 kV substation in the Maryland area. The request was received on December 17, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 660 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$245,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Barling / Claremont 138.0 kV	Cloverleaf / Fullerton Substation 138.0 kV	103.4	454	28.8
Sharyland / Tombstone 345.0 kV	Tombstone / Bixby Hill 500.0 kV	101.9	3,552	11.2
Sharyland / Bixby Hill 345.0 kV	Tombstone / Bixby Hill 500.0 kV L	100.4	3,496	11.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Claremont / Barling 138.0 kV	Reconductor line	\$1,405,000	7% (\$98,000)
Sharyland / Tombstone 345.0 kV	Reconductor line	\$16,692,000	0% (\$73,000)
Sharyland / Bixby Hill 345.0 kV	Reconductor line	\$15,041,000	0% (\$74,000)
	Total allocated to project		\$245,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.